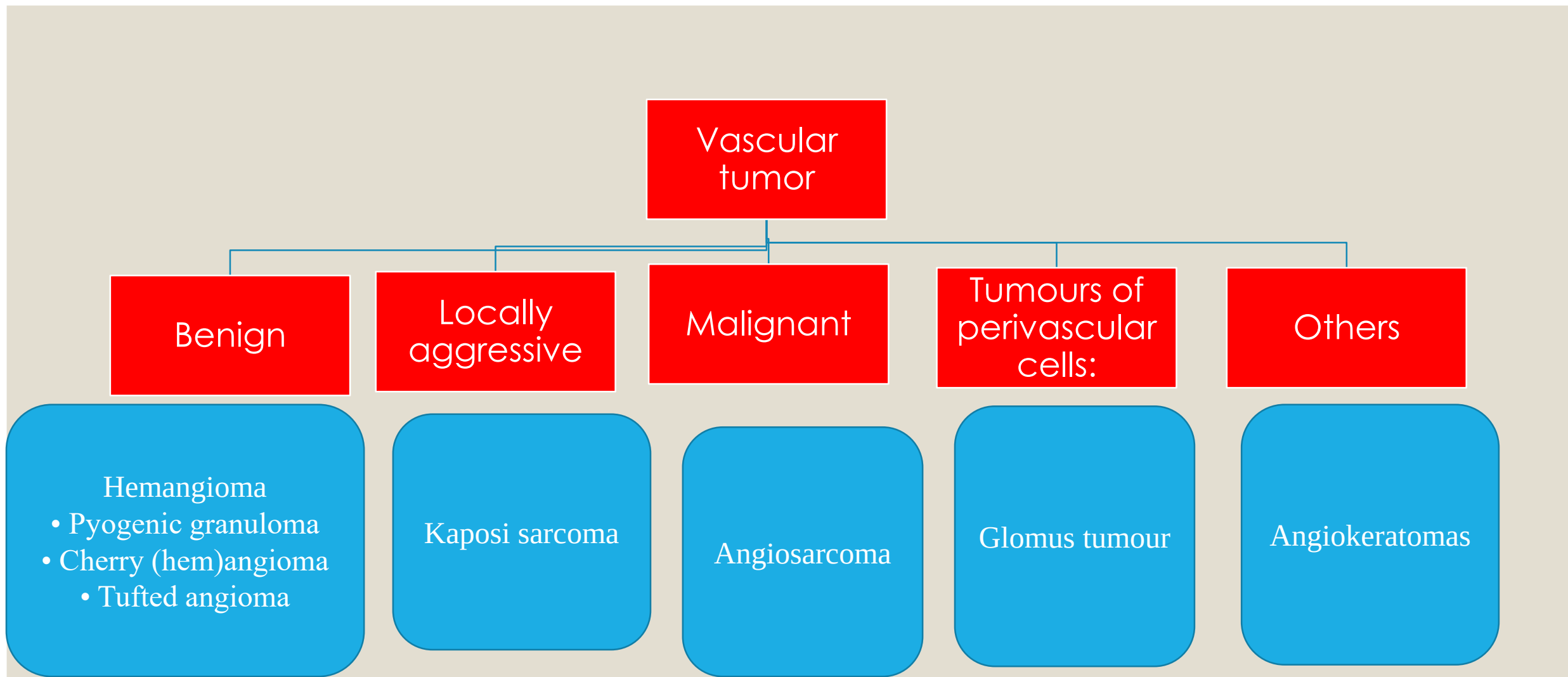


3-VASCULAR TUMORS ERYTHEMAS

Dr eman gomaa
Consultant dermatologist



1- Pyogenic granuloma

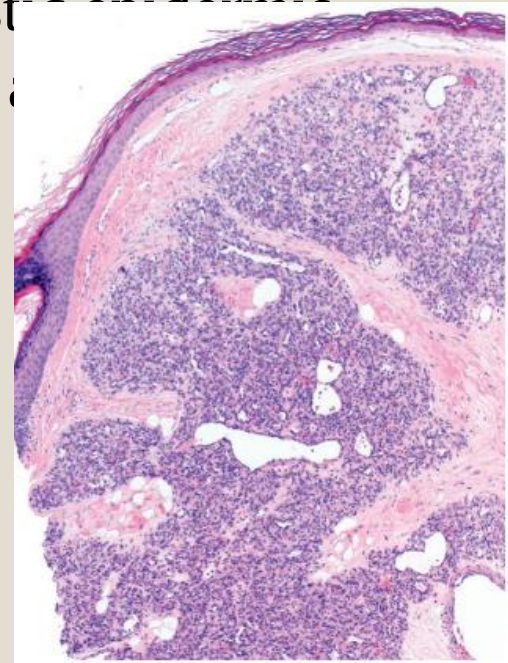
Lobular capillary hemangioma

- children and young adults
- reactive neovascularization process secondary to a injury or irritation
- solitary red papule that grows rapidly . It is extremely friable (ulcerates and bleeds profusely with minor trauma).
- The base is pedunculated and surrounded by a collar of acanthotic epidermis
- gingiva, fingers, lips, face and tongue.



- Histopathology

- Epidermis: atrophic or ulcerated.
- exophytic lesion with collarette of hyperplastic epidermis
- proliferation of small capillaries arranged in a lobular pattern

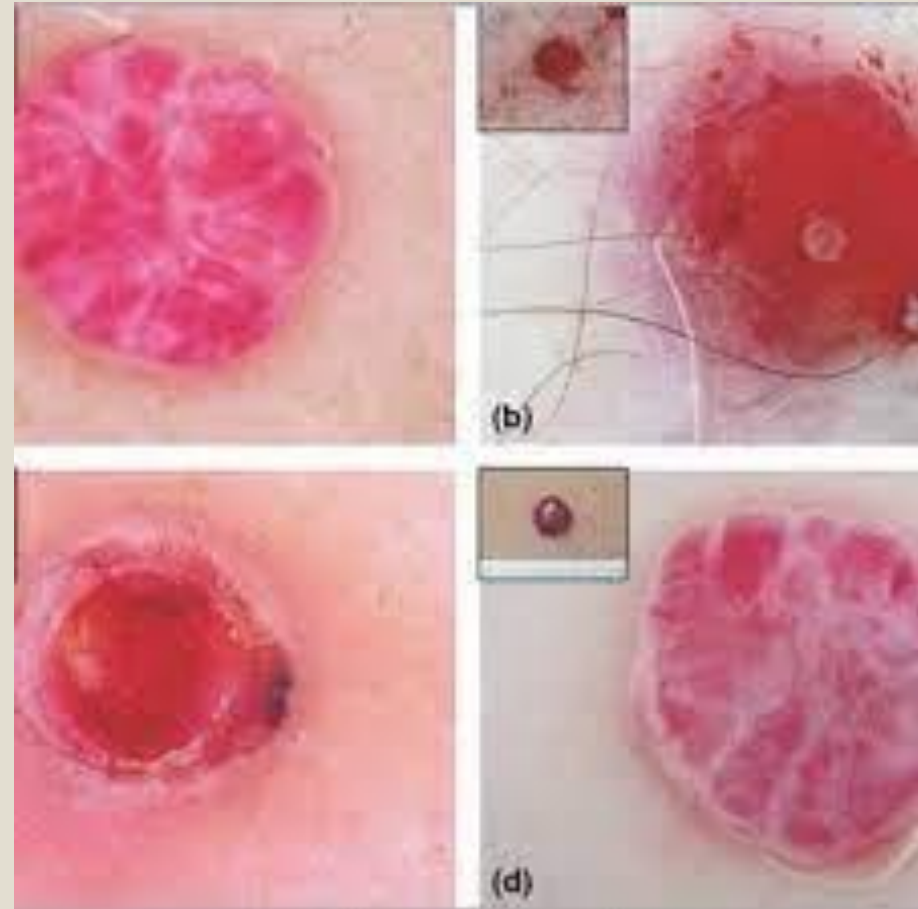


- DD
 - Hemangiomas
- • Bacillary angiomatosis
- • Spitz nevus
- • Kaposi sarcoma
- • Glomus tumors



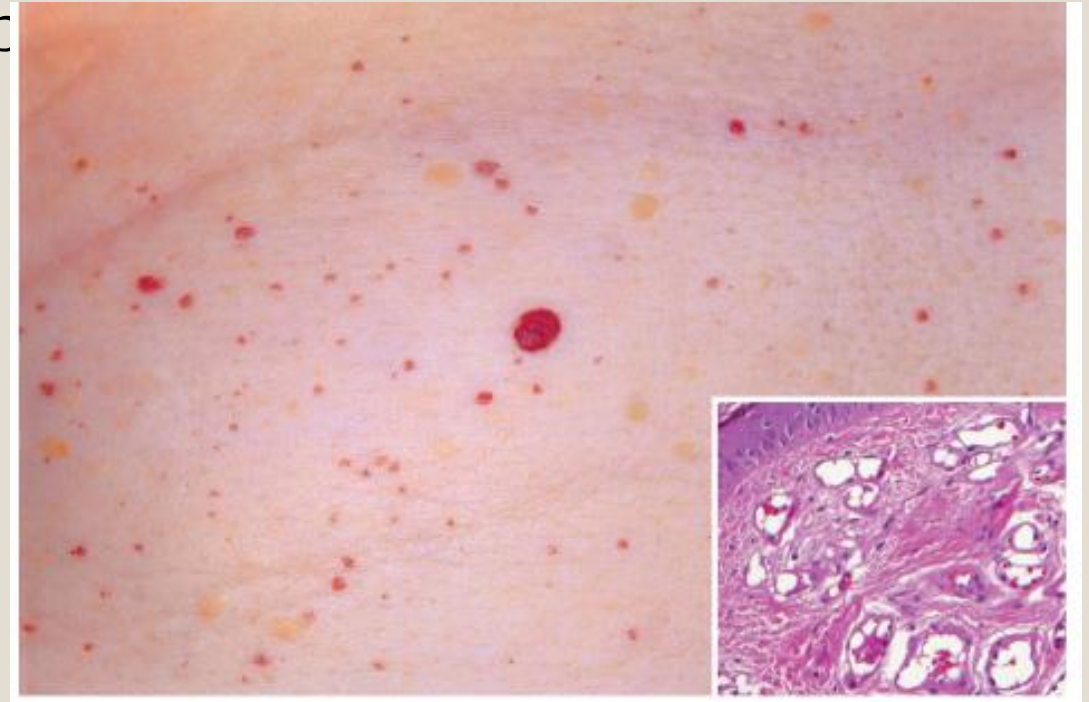
Treatment

- Shave excision
- electrosurgery .
 - Pulsed dye lasers
 - Sclerotherapy



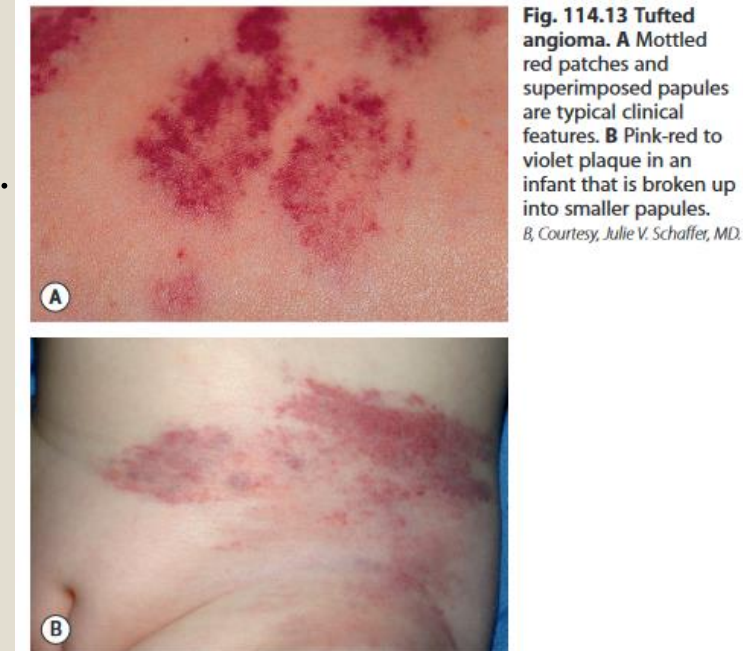
Cherry angioma

- A common, benign lesion found in most individuals by the age of 60 years, often in large numbers
- trunk
- Bright red dome shaped papules
-



Tufted angioma

- acquired lesion in children and young adults.
 - May be congenital (in this case, may be associated with Kasabach-Merritt)
- mottled red patches or plaques with superimposed papules, typically on the neck, trunk or shoulders
- histologically by tightly packed tufts of capillaries “cannonball” pattern.
- Complete surgical excision



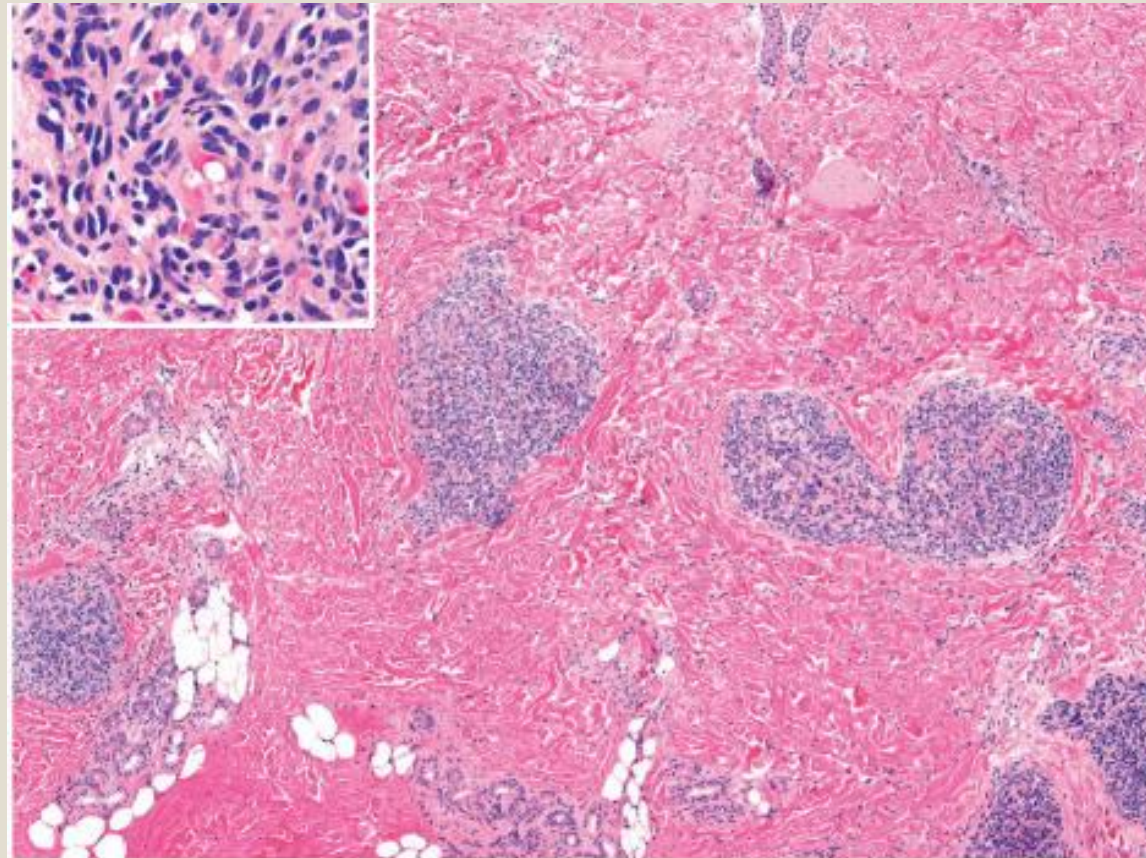


Fig. 114.14 Histology of tufted angioma. Within the dermis, there is typically a cannonball distribution of dense capillary lobules (inset). *Courtesy, Luis Requena, MD.*

Kaposi sarcoma

- multi-focal vascular neoplasm affecting the skin and other organs
- it is controversial whether Kaposi sarcoma (KS) represents neoplasia or hyperplasia; all clinical variants are viewed as a virally induced disease, with **human herpesvirus-8 (HHV-8)**
- HHV-8 contains homologues of cellular genes that can stimulate cell proliferation, inflammation and angiogenesis and inhibit apoptosis.



Variants of Kaposi sarcoma

- **1-Classic type**

- Commonly affecting distal parts of the lower limbs (feet) and occasionally hands, ears or nose.
- slowly-growing dark blue, purple or dark brown that pass into 3 stages: **patch / macule, plaque, and nodule (tumor)**
- Extra-cutaneous lesions: (10% of cases)
Lymph nodes, mucosae and viscera may be affected.
small intestine is the most common site



Fig. 114.24 Classic Kaposi sarcoma. Red macules and patches on the plantar surface of the foot, with violaceous plaques on the ankle.



EPIDEMIOLOGIC ASPECTS OF THE FOUR MAJOR TYPES OF KAPOSI SARCOMA

Classic

- Predominantly in individuals of Jewish (Ashkenazi) descent and/or of Mediterranean/Eastern European descent
- Male:female (M:F) ratio varies from 15:1 (older literature) to 3:1 or 1:1 (more recent observations)
- Two-thirds of patients develop disease after the age of 50 years

African endemic

- Incidence amongst black Africans ranges from 1% to 10%, with a clear male predominance
- Accounts for up to 9% of all reported cancers in equatorial Africa
- The lymphadenopathic variant affects primarily children, with a M:F ratio of approx. 1:1

Due to iatrogenic immunosuppression

- Individuals are immunosuppressed secondary to systemic medications, e.g. calcineurin inhibitors, prednisone, cytotoxic agents
- Cyclosporine is associated with a higher incidence and more rapid onset of disease
- Lesions can regress following cessation of immunosuppressive therapy
- Estimated incidences of post-transplantation KS in Western countries and the Near East are <1% and 4%, respectively
- Men are slightly more often affected than women

AIDS-related epidemic

- Occurs primarily in men who have sex with men
- To a lesser degree, seen in female partners of affected men and in women from certain regions of Africa and the Caribbean who became HIV-infected via heterosexual contact
- Up to 40% of men who have AIDS and became HIV-infected via homosexual contact can develop KS, compared with <5% in other risk groups and 4% in children
- Can flare in the setting of immune reconstitution inflammatory syndrome
- Occasionally develops in HIV-infected patients receiving chronic antiretroviral therapy, possibly due to immunosenescence

- *Endemic Africans KS*
- Cutaneous lesions: there are 4 variants:
 - Benign nodular:
 - Aggressive
 - Florid
 - Lymphadenopathic: more in children, equal sex *aggressive and fatal*

- **Epidemic (AIDS-related) KS :**
- Typically affecting the upper body, mid-face particular the nose
- gingiva and hard palate are also involved)
- lymph nodes, gastrointestinal tract , lungs



Actas Dermosifiliogr. 2018;109:878-87

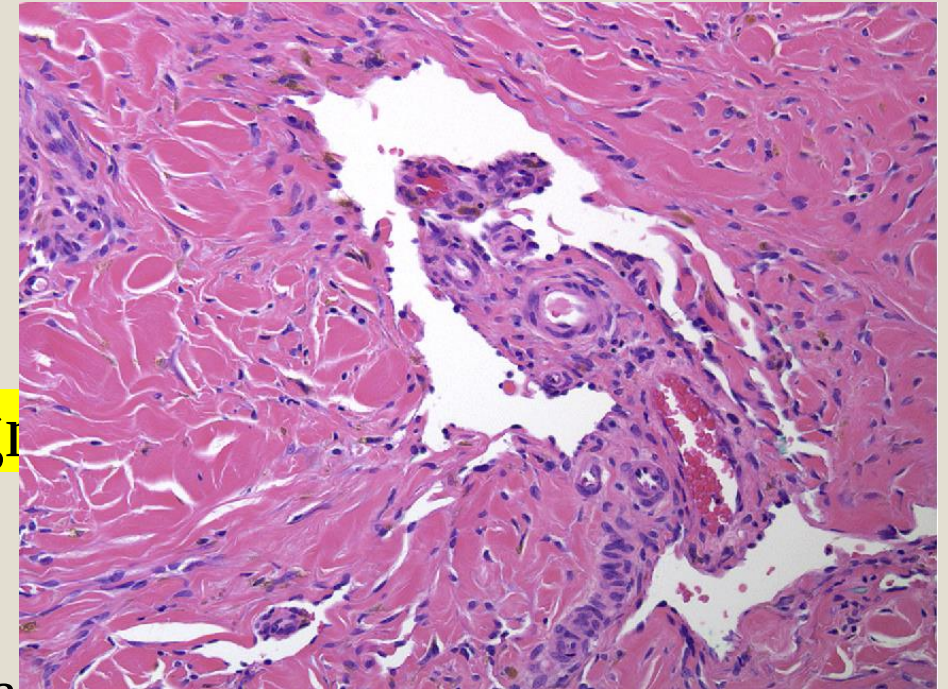


Fig. 114.25 Kaposi sarcoma in a patient with AIDS. Violet-red papules or nodules are often oval to lanceolate and are usually more widely distributed than in classic KS.

- ***4-Iatrogenic (Immunosuppression-associated) KS:***
 - prolonged immunosuppression e.g. after organ transplants
especially renal secondary to systemic medications Cyclosporine

Histopathology

- **Patch stage**: (mainly upper dermis)
 - Vascular spaces:
 - a few small thin-walled irregular / angulated spaces lined by endothelial cells that separate collagen bundles (slit-like).
 - pre-existing blood vessels (and normal adnexal structures) often protrude into newly formed blood vessels (promontory sign)
 - extravasation of RBCs
 - Cellular infiltrate:
 - patchy, sparse, peri-vascular infiltrate of lymphocytes and plasma cells.



- **Plaque stage: (whole dermis)**

- Vascular spaces: increase in number and acquire thicker walls.
- Cellular infiltrate: spindle-shaped cells with pink cytoplasm

- **Tumor stage: (whole dermis)**

- Increased spindle-shaped cells (replace dermal collagen, forming intersecting fascicles separated by vascular slits).

- **IMMUNOHISTOCHEMISTRY**

vascular markers (CD34, CD31, factor 8 related antigen).

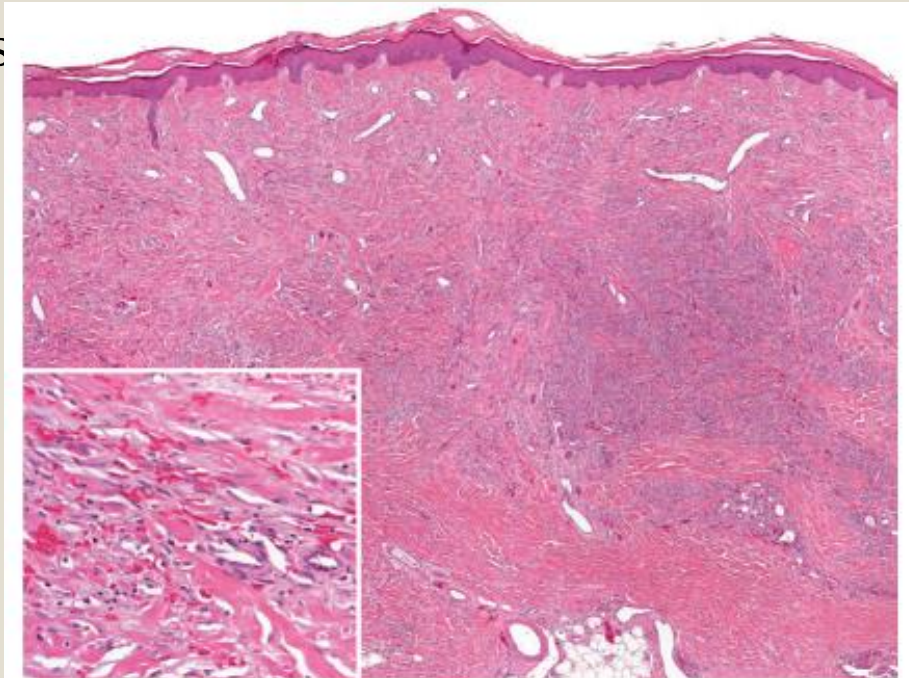


Fig. 114.26 Histology of Kaposi sarcoma – plaque stage. In the reticular dermis, there is a spindle cell vascular proliferation characterized by slit-like vascular spaces with extravasated erythrocytes (inset). Courtesy, Lorenzo Cerroni, MD.

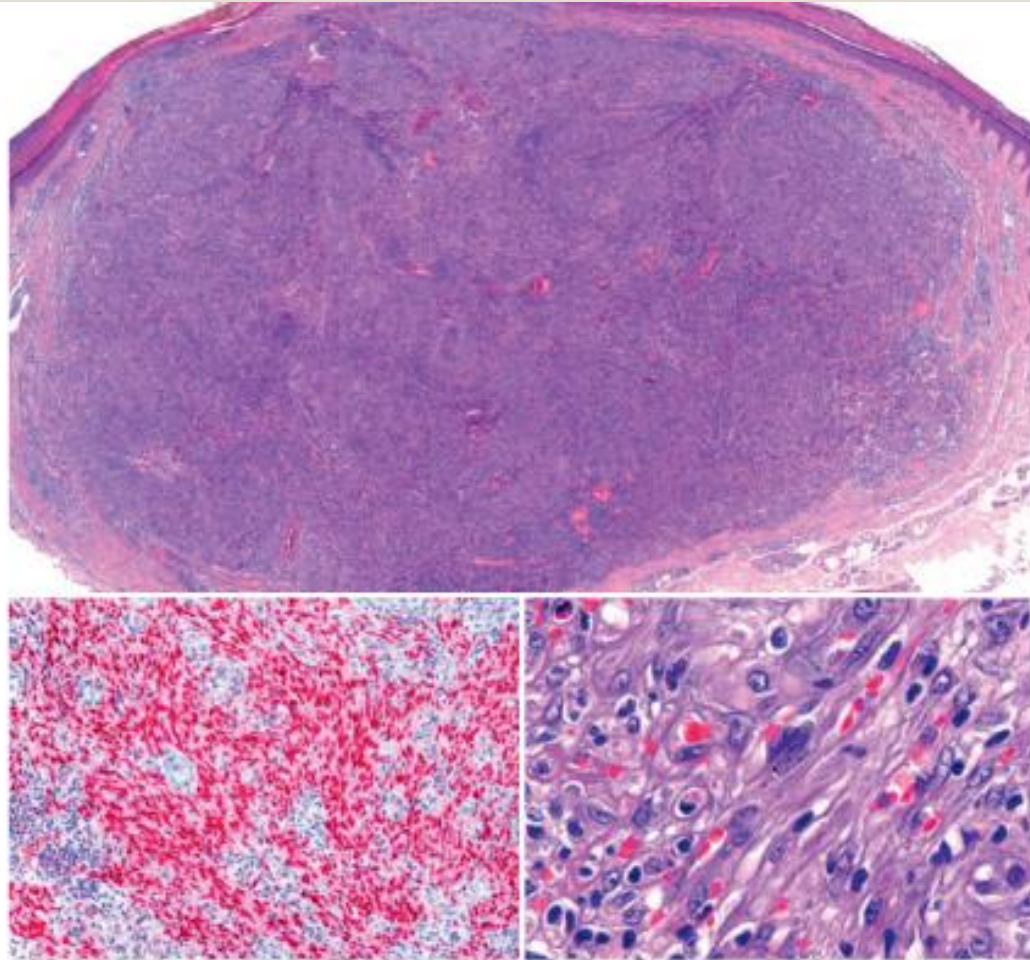


Fig. 114.27 Histology of Kaposi sarcoma – nodular stage. Nodular dermal proliferation of spindled and epithelioid cells admixed with extravasated erythrocytes (right inset). Note the positivity for the latency-associated nuclear antigen (LANA-1) of HHV-8 (left inset). *Courtesy, Luis Requena, MD.*

DD

Capillary malformation (patch stage)

- Hemangiomas (tumor stage)
- Angiosarcoma
- Cutaneous metastases or tumors (leukemia / lymphoma cutis)
- Venous and lymphatic malformations
- Acroangiokeratosis (pseudo-Kaposi sarcoma)

Treatment

- Treatment of KS depends on the clinical type and extent of involvement.

AIDS-associated KS: HAART regimen for treatment of HIV infection (often results in regression of KS).

- Iatrogenic KS: cessation or reduction of immunosuppressive drug

localized therapies: (used for localized lesions)

- Surgery (solitary lesions),
Cryotherapy
Laser surgery (PDL or CO2),
Photodynamic therapy,
Topical alitretinoin

- Systemic therapies: (used for extensive, multifocal disease).
 - Radiotherapy (multifocal but relatively localized KS).
- Systemic chemotherapy
 - Doxorubicin, vincristine, etoposide and/or bleomycin
- Interferon- α
 - drugs that inhibit angiogenesis or target HHV-8 e.g. thalidomide, bevacizumab (anti-VEGF)
 - matrix metalloproteinase inhibitors
 - tyrosine kinase inhibitors (imatinib, sorafenib, gefitinib)

Pseudo-Kaposi sarcoma

Acroangiiodermatitis of Mali

- Pseudo-KS is a benign reactive vascular hyperplastic condition
- five possible causes:
 1. Chronic venous hypertension (varicosities)
 2. Arterio-venous malformations of the lower limbs (Stewart-Bluefarb syndrome)
 3. Iatrogenic arterio-venous shunts
 4. Amputated limb
 5. Paralyzed limb



- more common in men.
 - Site: commonly occurs on the lower legs.
- minute purpuric macules that coalesce to form irregular plaques,
- signs of venous hypertension may be present (oedema, sclerosis, ulcers).
- **Histopathology :**
 - Benign capillary proliferations with plump endothelium
 - Extravasation of RBCs
 - NO spindle cell proliferation



DD

- KS
 - Gravitational eczema
 - Pigmented purpuric dermatosis (Schamberg's disease)
- Treatment
- Tt of cause
- Vascular support

	KS	Pseudo-KS
Vascular proliferations	Lie back-to-back	Separated from each other by edematous stroma
Vascular hyperplasia	Independent (NEW vessel formation)	Hyperplasia of pre-existing vasculature
CD34 staining	Endothelial lining and perivascular spindle cells	Endothelial lining ONLY

Angiosarcoma

- Uncommon malignant vascular tumor (<1% of all sarcomas).
- **Three types:**
 - (1) Angiosarcoma of the scalp and face of the elderly
 - (2) Angiosarcoma in areas of chronic lymphedema e.g. after radical mastectomy in ipsilateral arm (lymphangiosarcoma)
 - (3) Angiosarcoma following radiotherapy (high doses) occurring in areas of radiodermatitis
- Prognosis: extremely poor (< 15% of patients surviving 5 years)



Fig. 114.30 Post-radiation angiosarcoma. Red-violet patches and plaques on the breast. The clinicopathologic differential diagnosis is primarily post-radiation atypical vascular lesions of the skin, an entity that may represent a precursor of post-radiation angiosarcoma. Development of thicker, more indurated plaques and the presence of cellular atypia, multilayering of the vessels, and positivity for c-MYC favor angiosarcoma.



well to moderately differentiated lesions composed of anastomosing vessels that dissect between bundles of collagen. Approximately 50% of cases express markers of lymphatic differentiation.

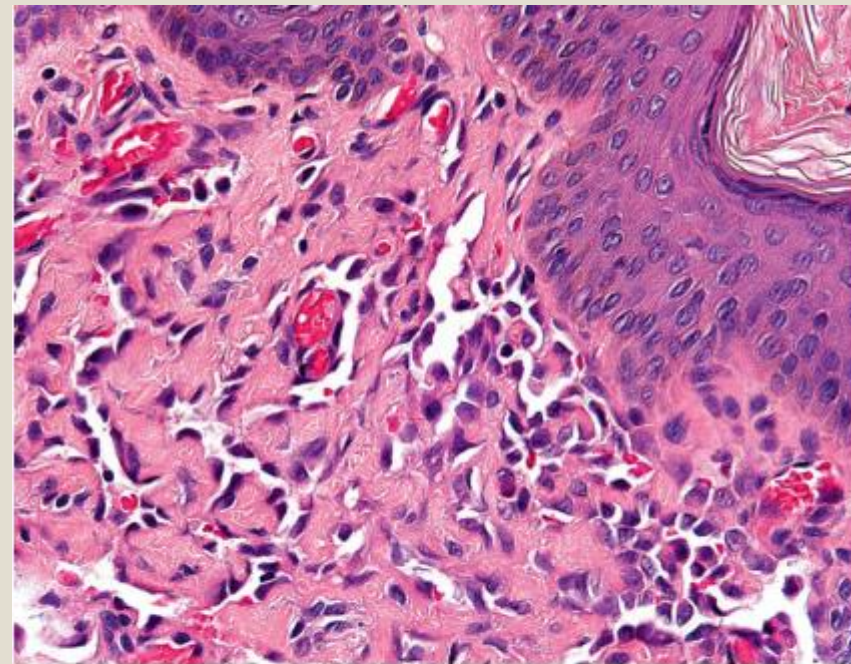


Fig. 114.32 Histology of angiosarcoma with lymphatic characteristics ("lymphangiosarcoma"). Some of the vascular channels are angular and lack erythrocytes. Also note the hyperchromatic atypical endothelial

Benign vascular neoplasms and neoplastic-like proliferations

- Infantile hemangioma (see Ch. 103 for details)
- Congenital hemangioma (includes non-involuting congenital hemangioma [NICH], rapidly involuting congenital hemangioma [RICH], and partially involuting congenital hemangioma; see Ch. 103 for details and mutated genes)
- **Pyogenic granuloma (lobular capillary hemangioma)***
- **Cherry angioma**
- **Tufted angioma (likely a superficial form of kaposiform hemangioendothelioma)**
- **Glomeruloid hemangioma***
- **Microvenular hemangioma***
- **Epithelioid hemangioma (angiolymphoid hyperplasia with eosinophilia [ALHE])***
- **Multifocal lymphangioendotheliomatosis with thrombocytopenia[#]**
- **Sinusoidal hemangioma****
- **Spindle cell hemangioma*****
- Epithelioid angiomatous nodule (debate as to whether it is a variant of epithelioid hemangioma)
- Papillary hemangioma
- Acquired elastotic hemangioma (capillary-like appearance in association with solar elastosis)
- Cutaneous pseudoangiomatous stromal hyperplasia (primarily encountered in the breast)
- Post-radiation atypical vascular lesions of the skin

Borderline and low-grade malignant vascular neoplasms

- **Kaposiform hemangioendothelioma (KHE)**
- **Papillary intralymphatic angioendothelioma (PILA) (Dabska-type hemangioendothelioma, Dabska tumor)**
- **Retiform hemangioendothelioma (RHE)**
- Epithelioid sarcoma-like hemangioendothelioma (pseudomyogenic hemangioendothelioma)
- **Kaposi sarcoma**

Malignant vascular neoplasms

- **Epithelioid hemangioendothelioma (EHE)[†]**
- **Angiosarcoma**

Perivascular neoplasms and neoplastic-like proliferations

- **Glomus tumor (proper)**
- **Glomuvenous malformation ("glomangioma")**
- **Glomangiosarcoma**
- **Infantile-type hemangiopericytoma/myofibromatosis**
- Adult-type hemangiopericytoma
- Glomangiopericytoma
- Myopericytoma
- **PEComa (perivascular epithelioid cell tumor)[†]**

Glomus tumor

- A tumor arising from the myoarterial glomus cells
- Glomus tumors are composed mainly of vascular channels surrounded by proliferating glomus cells.
- 1-Solitary glomus tumor: solid (glomus cells > vascular spaces)
2. Multiple glomus tumors: angiomatous (vascular spaces > glomus



Fig. 114.33 Glomus tumor. The lesion presented with pain and ill-defined subungual erythema. *Courtesy, Ronald P Rapini, MD.*



Fig. 114.35 Glomuvenous malformation ("glomangioma"). Note the cluster of slightly compressible blue papules.

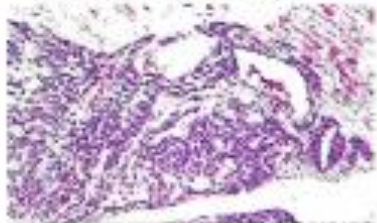
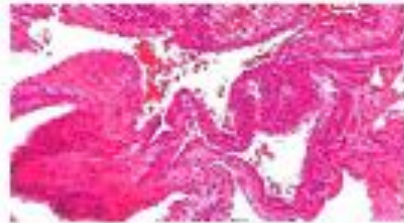
	Solitary glomus	Multiple glomus
Epidemiology		
Incidence/prevalence	Relatively uncommon .	
Age	More common in adults (3 rd – 4 th decade).	10 times more frequent in children than in adults.
Sex	Equal sex distribution.	
Familial cases	Not reported.	May be present (see later)
Clinical features		
Site	The commonest site is the hands , particularly the fingers (nail beds) , followed by toes, other sites on the extremities and other body regions.	Less restricted to the extremities, may affect any area of the body and may be generalized.

	Solitary glomus	Multiple glomus
Symptoms	Paroxysmal pain (provoked by direct pressure / change in skin temperature or spontaneous)	Not usually painful (grouped multiple tumours may be painful).
Lesions	A pink or purple nodule of variable size. 	Larger lesions, usually dark blue in colour. 



Fig. 114.36 Glomuvenous malformation ("glomangioma"). Large cluster of blue papulonodules becoming confluent centrally.

Histopathology

	Well-circumscribed, lobulated, capsulated dermal mass.	Ill-defined and NON-capsulated.
	Several narrow vascular lumina lined with a single layer of endothelial cells.	More numerous vascular lumina lined with a single layer of endothelial cells.
	Surrounded by multiple layers of glomus cells	Fewer number of glomus cells
	Cellular (glomus cells > vascular spaces) 	Angiomatous (vascular spaces > glomus cells) 
Glomus cells	– Monotonous cuboidal, with a well-marked cell membrane, faint eosinophilic cytoplasm and a round central dark nucleus (epithelioid). – These cells are aligned in rows around the single layer of endothelial cells of the vascular spaces. – Electron microscopy suggests that glomus cells are transversely cut smooth muscle cells.	
Immunohistochemistry	Tumour cells are universally positive for smooth muscle actin and are usually <i>negative for desmin</i> (SGE Dermatology / EBDM by Dr Emma Connolly Academy) <i>(positive in majority of cases). Positivity for CD34 may be seen.</i>	

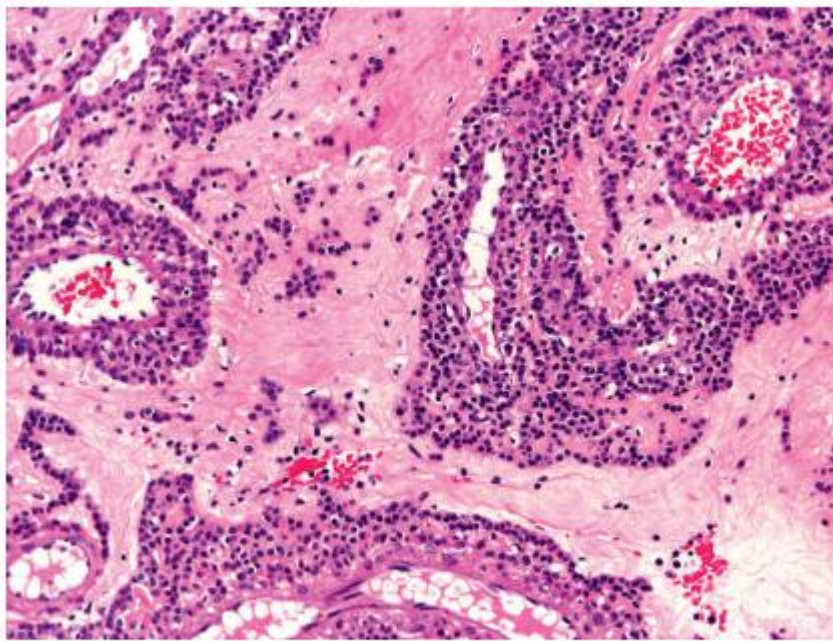


Fig. 114.37 Histology of glomus tumor. Proliferation of rounded, homogeneous, hyperchromatic glomus cells surrounding vascular lumina.
Courtesy, Lorenzo Cerroni, MD.

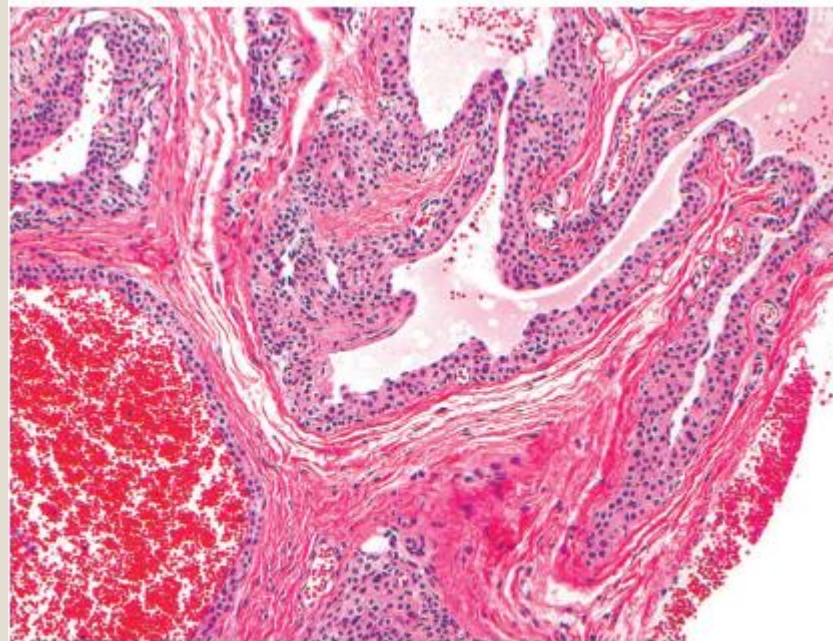


Fig. 114.38 Histology of glomuvenous malformation ("glomangioma"). One or more, often incomplete, layers of glomus cells rim abnormal venous channels that are otherwise histologically consistent with a venous malformation.

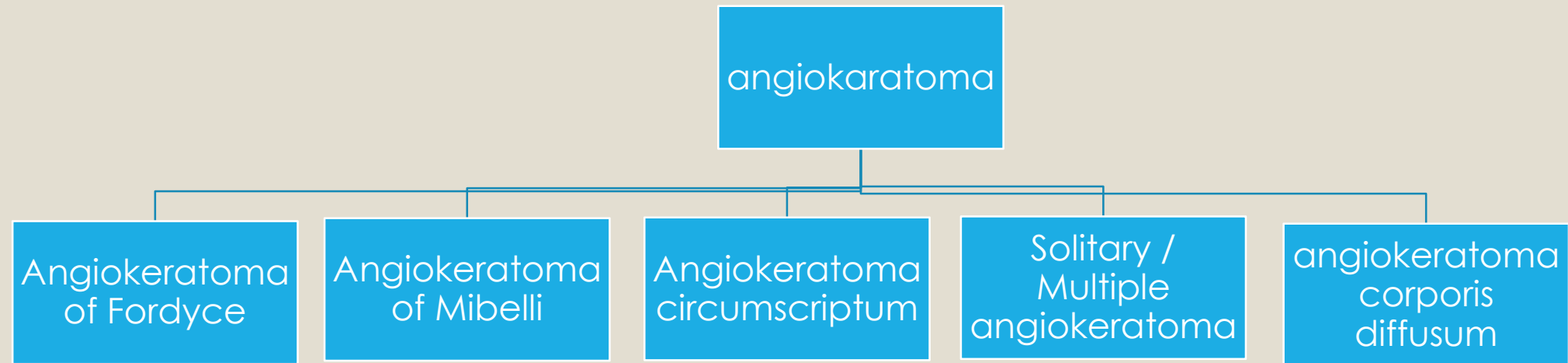
	Solitary glomus	Multiple glomus
Differential diagnosis		
	Other painful skin tumors (see below).	Cavernous haemangioma, venous malformations, blue rubber bleb naevus
Treatment		
– Surgical excision, electrocautery or sclerotherapy. – In multiple tumors, these modalities are difficult or less effective. Vascular lasers (PDL, Nd:YAG, KTP) are a good choice for multiple, as well as, solitary lesions. – Local recurrence is very rare (related mainly to incomplete excision or deeper lesions with an infiltrative growth pattern described as '<i>infiltrating glomus tumours</i>').		

painful skin tumors (LEND AN EGG):

Leiomyoma,
 Eccrine spiradenoma,
 Neuroma,
 Dercum's disease,
 Angiolipoma,
 Neurilemmoma
 (Schwannoma),
 Endometrioma
 Glomus tumor (solitary)
 Granular cell tumor

Angiokeratoma

- Angiokeratomas are small benign vascular skin lesions
vascular dilatation 'ectasia' (angio-) and
acanthosis / hyperkeratosis (-keratoma)
Angiokeratomas are not true angiomas (tumors) as they occur in pre
existing vessel



1-Angiokeratomas of Fordyce

- second or third decade
- scrotum
- associated with varicoceles
- Vulvar lesions may be associated with vulvar varicosities,
- hemorrhoids, oral contraceptive use



2) Angiokeratoma of Mibelli

- Children and young adults
- dorsal and lateral aspects of the fingers, toes, hands and feet
- Multiple, bilateral and symmetrical minute bright red macules that , becomes elevated warty



3-Angiokeratom a circumscriptum

- May be congenital (present at birth) or acquired
- females
 - A plaque of multiple discrete papules
- trunk and extremities mostly unilateral
- Associated diseases:
- , Klippel-Trenaunay syndrome(KTS)
- , naevus flammeus,
- cavernous hemangiomas



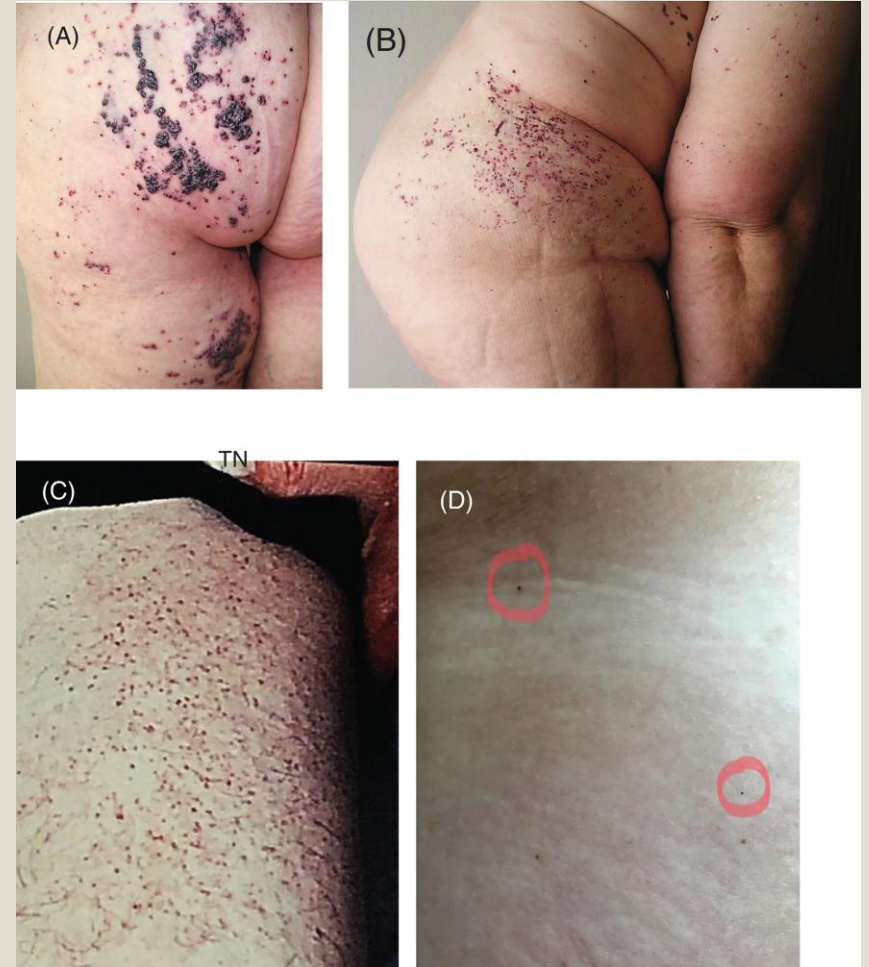
Solitary / Multiple angiokeratoma:

- Blue-black small warty papule(s)
 - Any site, most commonly lower limbs

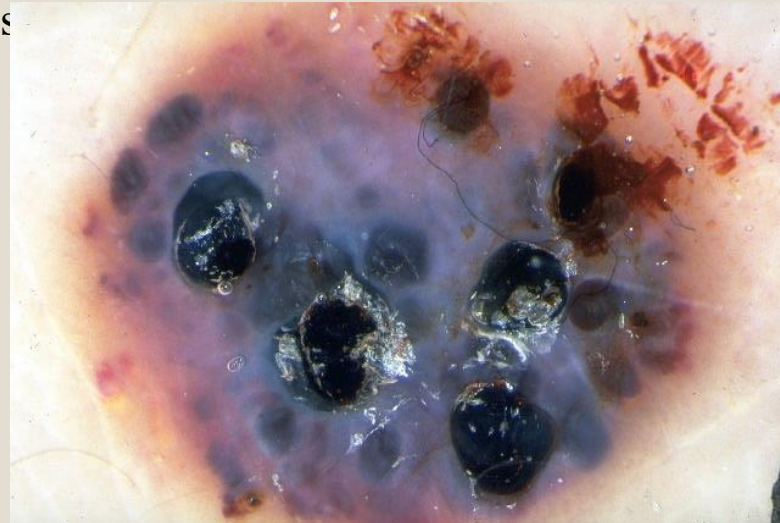


Angiokeratoma corporis diffusum

- skin manifestations of Fabry disease.
- angiokeratomas, usually in a bathing trunk distribution
- NB:
- **Fabry disease:**
 - X-linked recessive sphingolipidosis
- Deficiency of the lysosomal enzyme α -galactosidase
- corneal opacity
- paraesthesia,
- hypertension,
- renal affection



- H.P
- hyperkeratosis,acanthosis, papillomatosis and elongation of rete ridges.
 - marked dilatation of blood vessels
- o treatment is required for asymptomatic lesions.
 - Removal for cosmetic reasons



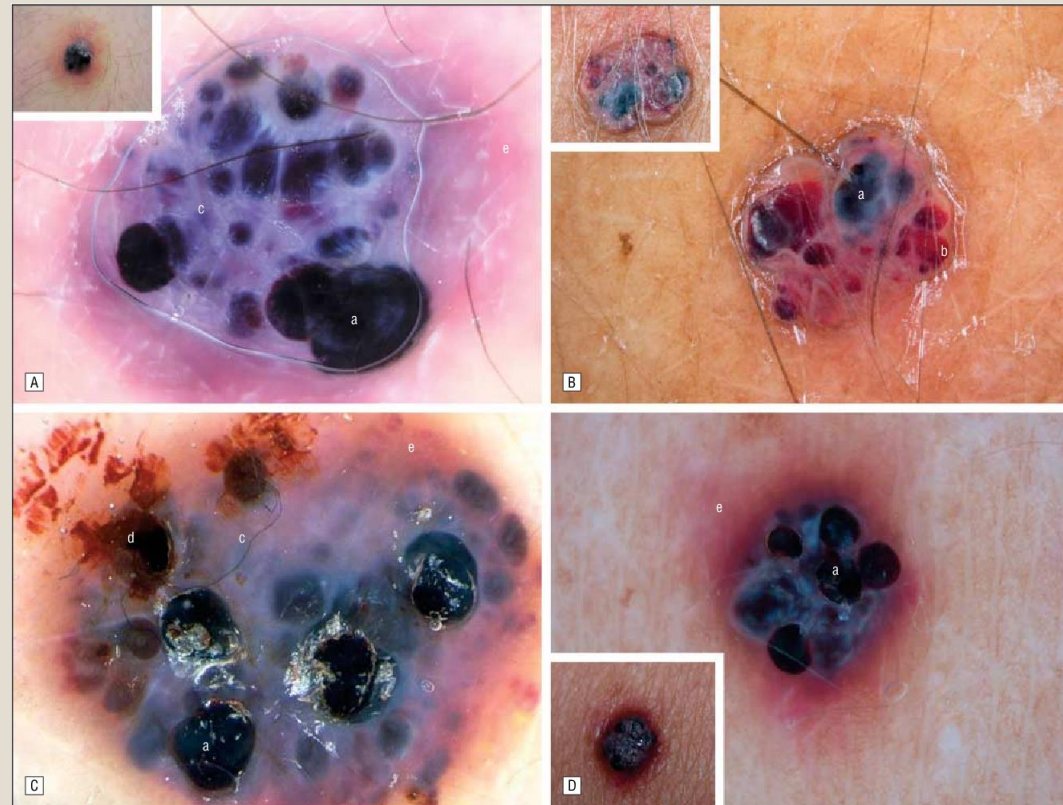


Figure 1 Dermoscopic structures of solitary annular keratomas. Located on the left leg of a 28-year-old man (A), the right leg of a 27-year-old man (B), the trunk of a

Angiolymphoid Hyperplasia with Eosinophilia

- A benign locally-proliferating vascular lesion.
- AHE is a reactive, not a neoplastic.
 - Trauma
 - AV shunting
- multiple clustered small papules or nodules, brown, pink in color
- head and neck (especially around the ears)
- Extra-cutaneous manifestations:
Regional lymph node enlargement
- Eosinophilia
- Bone affection



Figure 1 Dome-shaped, dull erythematous,

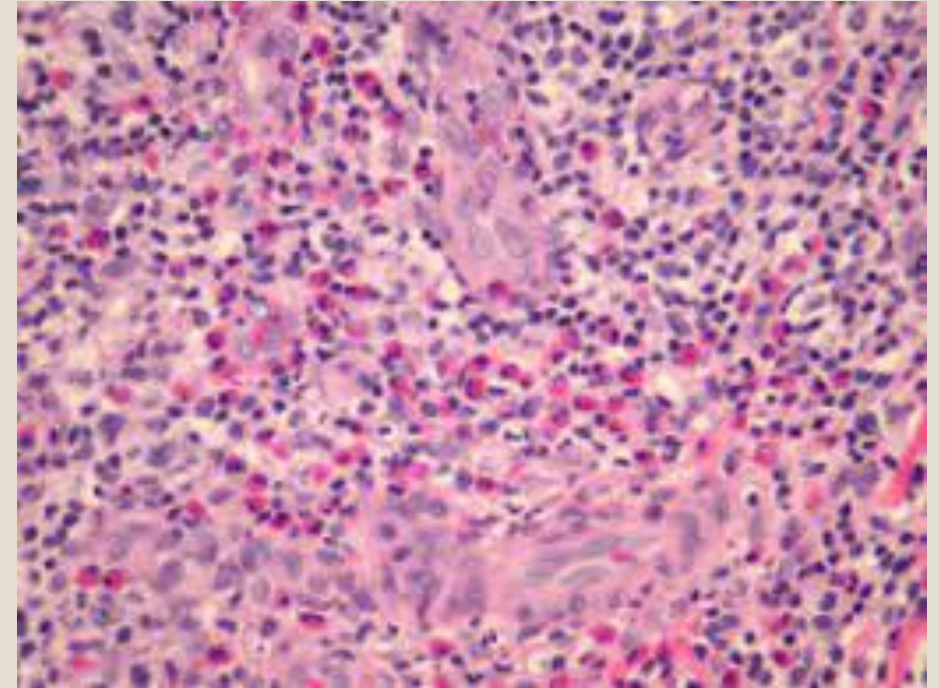
- Vascular proliferations lined by enlarged plump (hobnail) endothelial that protrude into the lumen —> scalloped or “cobblestone”
- Eosinophilis
- DD
- Pseudo-lymphoma
 - Sarcoidosis
 - Angiosarcoma
 - metastatic tumors
 - Kimura’s disease:



TREATMENT:

Surgical excision (one-third of lesions recur)

- Lasers (e.g. CO₂, PDL, Nd:YAG)
- Sclerotherapy
- Radiotherapy
- Imiquimod cream





Thank you